
Limited by Data: A Tool for Insight-driven Forecasting of County-level Yield

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Abstract

Many parties, both in academia and industry, need reliable information regarding crop yield. The primary goal of this project is to forecast corn yield production in the U.S. Corn Belt using a variety of modeling techniques. Sometimes, machine learning methods yield results that are hard to interpret. So, the second objective is to visualize yield and climate variables, while also allowing users to build their own forecasts. This dataset features meteorological data for 1,400 counties in the Corn Belt, as well as a final yield output in bushels per acre. For each county, 39 years of entries are present. In terms of modeling, we surprisingly found that the linear models performed better than the time-series models. We were also able to build a user-friendly tool that allows stakeholders from all fields to explore the data and make their own forecasts ¹. So, these results are useful in that we not only tuned our own models, but also created an application that lets non-technical parties make yield predictions with personalized feature values.

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¹Dashboard Link

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1 Introduction

Corn yield forecasts guide the decision-making of a plethora of stakeholders. Government agencies need emergency plans, farmers want realistic expectations, and traders rely on accurate predictions. With increasing climate unpredictability, implementing sound predictive models becomes a critical task. Coupled with this is the need for effective data communication in terms of machine learning explainability. So, the first aim of this project is to construct linear and nonlinear models that deliver reliable yield forecasts. The second aim of this project is to create a visual tool that allows for granularity of detail, which is otherwise not found in traditional machine learning projects.

To achieve these two objectives, we used daily data containing corn yield (bushels/acre) from counties in the US Corn Belt from 1980 to 2018. This was accompanied by meteorological data for these counties during the growing period (210 days). Some of these climate-related features data include precipitation, vapor pressure, and humidity. To synopsise this large quantity of data, we calculated the yearly averages of all these variables. However, for time-series tasks, we left the data as is.

In terms of forecasting, we incorporated a mix of classical linear methods, time-series techniques, and deep learning frameworks. We found that the linear models actually outperformed the time-series and deep learning models; however, these models struggled in predicting anomaly years with unusually high or low corn yield.

When focusing on explainability, we deemed it crucial to create a user-friendly interface. We utilized Streamlit to build an app that allows stakeholders to explore the data from a spatiotemporal lens. Users can view a county-level choropleth featuring any variable for a selected year. This app also includes model-building functionality and climate stress-testing capabilities. To further enhance model explainability, we crafted ALE plots that isolate the impacts of certain predictive features on yield predictions.

2 Prior Work

2.1 Background Research

Our project and data comes from Liu et al. (2022), which compares different ML-based algorithms against process-based models to evaluate how well they are able to predict corn yields across different years. They found that a deep learning model performed the best compared to traditional machine learning models or a process-based one. In addition we looked at You et al. (2017), which hypothesized that deep learning can learn more effective features from data than traditional learning techniques. We based our LSTM + Attention model off of the ideas of these two papers, but needed to look at other literature to figure out other classical regression models to use.

2.2 Classical Regression

We next looked at Leng and Hall (2020), which investigates the strengths and weaknesses of three different model types: process-based models, a traditional regression model, and a Random Forest model, also in predicting corn yield across the US. In simulating average yields, the regression and Random Forest models showed better results in reproducing observed spatial patterns of maize yield compared to process-based models. Hou et al. (2023) further explains that classical linear modeling is highly effective and remains the superior method over non-linear models for addressing real-world problems that have longitudinal data with random effects. They were able to get improved predictions using the Lasso and Ridge regression models. Thus, we chose to implement these models as well as Elastic Net regression to see how they would do with our corn yield data.

2.3 Time-Series

We wanted to use two different time-series modeling techniques: auto regression and deep learning. For auto regression, Zivot and Wang (2003) outlines the Vector Autoregressive framework for analyzing multivariate time series. The VAR framework is an extension of the Autoregressive model where $VAR(p)$ with lag-order p describes a vector of variables Y_t as a function of its own past values and an unobservable error term. We used the findings of this paper as a blueprint for the implementation of VAR modeling in crop yield data. We expanded on this research by adding first-order differencing to the feature and target variables before forecasting to achieve higher accuracy.

As mentioned previously, Liu et al. (2022) used a deep learning model called LSTM (Long Short Term Memory) and found that it performed the best. Siami-Namini et al. (2018) compares two different time series models: ARIMA (Auto Regressive Integrated Moving Average) and the LSTM. The goal was to see which model made the best predictions in terms of lower forecast errors and accurate predictions. They used financial data from Jan 1985 to Aug 2018 and found that the LSTM performed better than the ARIMA model (84% to 87% reduction in error rates). While our project does not use financial data, this paper was still useful to us as we wanted to test to see how the two models did on our data.

2.4 Distribution Regression

The last modeling effort that we did was distribution regression (DR). Lemerrier et al. (2021) contributes a framework for implementing DR models on sequential data from all spheres. The authors explored ways to apply this technique to multiple time series data. To tackle this problem, the group applied their techniques to financial, thermodynamic, and agricultural data.

This technique works relatively well in predicting a target from multiple data streams, and it is something that could be used in future work. It was not implemented in this project because the only way to give it a "distribution" of time series was by grouping counties together and then making predictions on this group. This would be a different target from our other models.

2.5 Visual Tools

Sorapure (2023) This paper analyzes the value of actionable dashboards in terms of user behavior. Interviewees were asked to navigate two dashboards with COVID-19-related information. The study revealed that personal interests and situations guided participants to different pages and individual insights. The key detail is that these interviewees were not domain experts.

This served as inspiration to build a dynamic dashboard that lends itself to a plethora of conclusions. Since our parties of interest possess a diverse array of backgrounds, it is crucial that a dashboard allows for interpretable, personal insights that hone in on respective regions, times, and variables of interest.

3 Data

3.1 About the Data

We gathered our dataset from two distinct sources:

1. GridMET, developed by Abatzoglou (2013) and downloaded from the Northwest Knowledge Network
2. US Department of Agriculture (USDA)

GridMET provides 16 different surface meteorological variables for the contiguous US at ~ 4 km resolution from the years 1979 to the present day. We used 10 out of those 16 variables, and the years 1980 – 2018. The primary features are related to temperature, precipitation, and sunlight:

- **sph**: Near-Surface Specific Humidity
- **vpd**: Mean Vapor Pressure Deficit
- **pr**: Precipitation
- **rmin**: Minimum Near-Surface Relative Humidity
- **rmax**: Maximum Near-Surface Relative Humidity
- **srad**: Surface Downwelling Solar Radiation
- **tmmn**: Minimum Near-Surface Air Temperature
- **tmmx**: Maximum Near-Surface Air Temperature
- **vs**: Wind speed at 10 m
- **th**: Wind direction at 10 m

The **Northwest Knowledge Network** provides research data management and is managed by the University of Idaho. It is a place for researchers to store their data for others to freely use. We were able to download the GridMET dataset from this website for the specified years in a `.npy` file format, which we could easily transform into a Pandas `DataFrame`.

The **USDA** provides the county-level annual corn yield, also for the years 1980 – 2018 in the form of a `.csv` file. We could also easily transform it into a Pandas `DataFrame` and then perform a join on the counties to combine the meteorological variables and the yield into one dataset.

The meteorological data was already cleaned so that it only included data for the counties that reported yield for any given year.

3.2 Data Formats

In the end, we had two formats of the data that we could work with:

1. A single file with averaged features variables and the yield values
2. Separate files for the different feature variables and another one for the yield

Regardless, the general format for both file types was the same. Each file has 45,338 rows where each row value represents a single county for a single year. For example, for 1980, there were 1438 counties that reported the yield. All of the files that we worked with were aligned on the rows, which made it easy to combine them as needed.

3.3 Data Analysis/Complex Visualization

Before doing any modeling, we performed exploratory data analysis to better understand what the data looked like. In Figure 1, we looked at the trend of corn yield over time. While it does look to increase over time, there are a few anomaly years where extreme weather events occurred such as 2012.

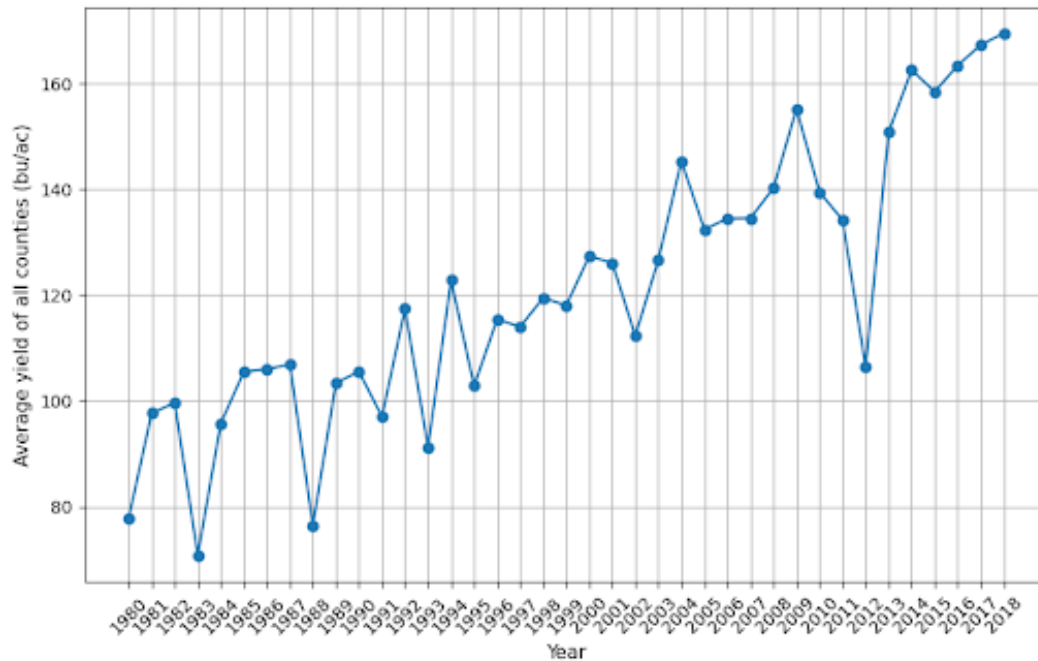


Figure 1: Average corn yield over the years 1980-2018, measured in bushels per acre.

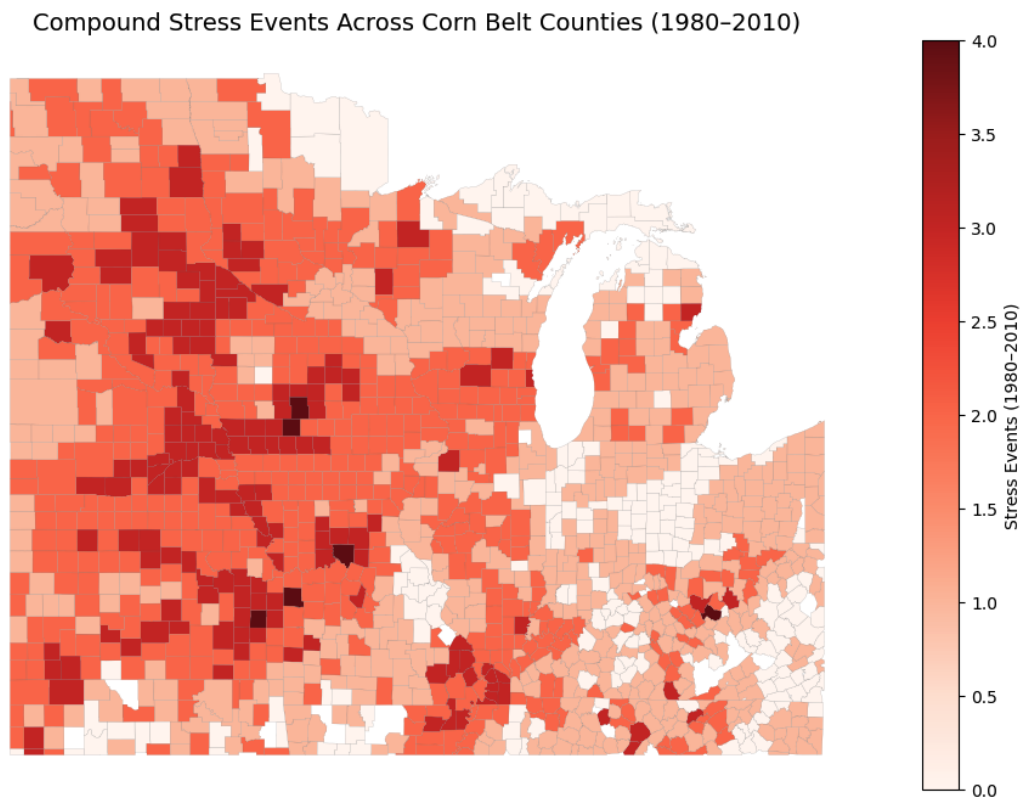


Figure 2: Choropleth map of compound stress events, where compound stress events are years where the value was either in the top/bottom 10%

Figure 2 illustrates counties experiencing a high frequency of compound stress events associated with reduced crop yields. Compound stress events are defined as years in which the annual mean of at least one meteorological variable falls within the upper or lower 10% of its observed distribution. We can see here that there are certain counties that experienced high-anomaly weather events.

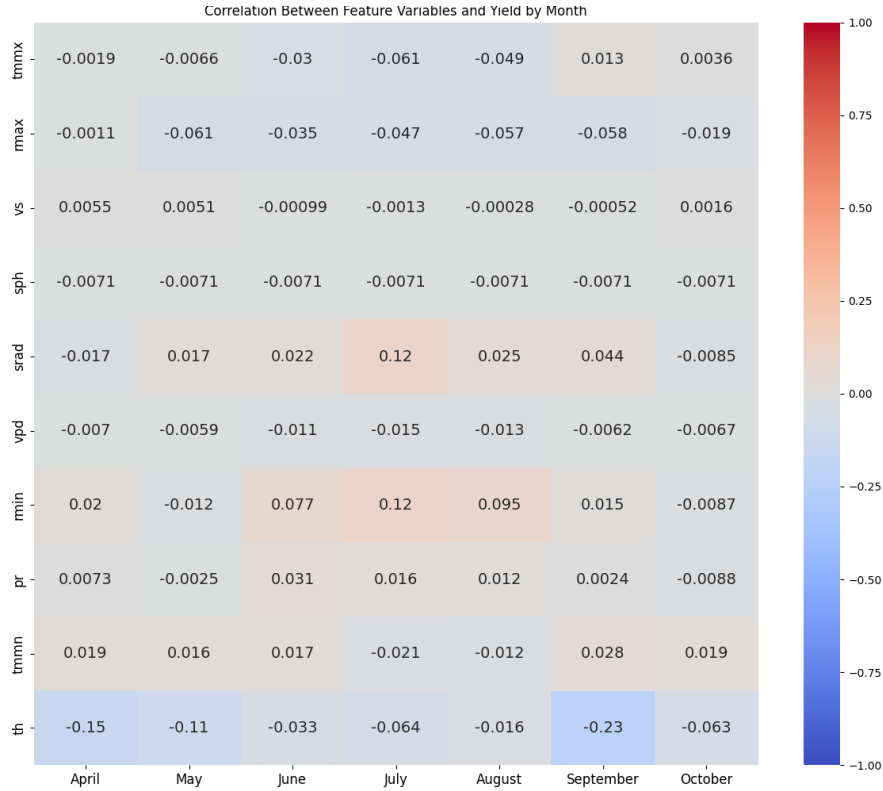


Figure 3: Correlation matrix between feature variables and yield by month

For each of the 10 variables described in Section 3.1, we created a correlation matrix between each of the variables and the yield per month of the growing season (Figure 3). Most of the variables had almost no correlation and a few had small negative and positive correlations. From this, we were able to see that there is no strong correlation between any one of the variables and yield.

Correlation between temperature and yield within counties for May - June

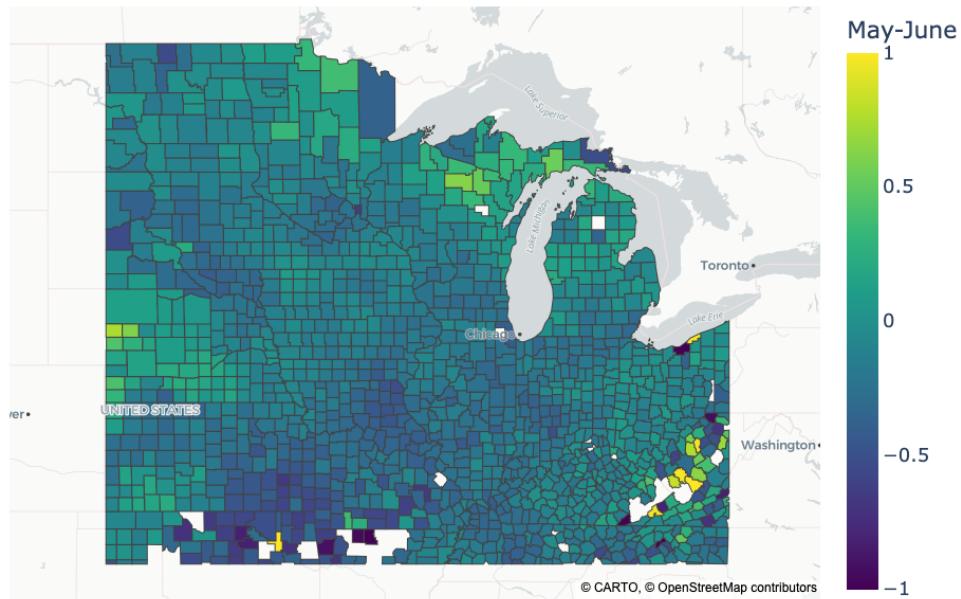


Figure 4: Choropleth map comparing the correlation between temperature and yield in the months of May and June. The values were averaged over all available years of data

Another correlation that we looked at was specifically between temperature and yield for the first few months of the growing period. Figure 4 shows how a majority of the counties have medium to weak correlation strength.

4 Machine Learning

4.1 Setup

The following data partitioning strategy, inputs and outputs, and evaluation metrics were used for all models.

4.1.1 Data partitioning

For all models, **leave-one-year-out cross-validation (LOYO)** was used. Leave-one-year-out is a cross validation strategy in which models are trained on all years except one and evaluated on the held-out year. This process is repeated so that each year serves as the test set (2005-2015) exactly once. For example, in the first round of cross-validation, a model is trained on the years 1989-2004 and then tested on year 2005. For the second round, a new model is trained on the years 1990-2005 and then tested on year 2006.

4.1.2 Inputs and Outputs

The inputs to the model are as described in Section 3.1. The outputs are yield predictions per county. Depending on the models, the meteorological variables are either averaged per year or used as-is.

4.1.3 Evaluation Metrics

Root mean squared error (RMSE) and mean absolute error (MAE) were calculated to assess deviation from actual yield values, and the coefficient of determination (R-squared) was calculated to assess how well the models fit the data.

4.2 Classical Regression Models

To perform the classical regression techniques, the features were aggregated for each county in each year by mean, standard deviation, minimum, maximum, first quartile, median quartile, third quartile, skew, kurtosis, previous year yield, and yield trend (rolling average over the previous three years). Ridge, Lasso, Elastic Net, and Support Vector Regression models were implemented and compared. For Ridge regression, alpha was set at 200.0. For Lasso regression, alpha was set at 0.15. For Elastic Net regression, alpha was set at 0.01, and the l1 ratio was 0.5. For support vector regression, a radial basis function kernel was used, C was set at 10, and epsilon was set at 0.1. The Lasso regression model performed the best, having the lowest RMSE and MAE values, as well as the highest R-squared value compared to the rest of the tested models (Table 2). Although the performance of these models was promising, when assessing the deviation metrics per test year, it is possible to see that the accuracy of prediction for anomaly years or years following anomaly years was worse. For example, the RMSE and MAE values when using 2012 as the test year were high 5, likely due to the unusually low yield during this year (as the result of a drought) 1. Additionally, the non-uniformity performance of 2013 vs 2012 is attributed to learning using previous 3-year yields, which causes 2013 to learn from 2012 yields. This non uniformity is something we plan to explore in future works. Overall, the classical regression models performed well in predicting crop yield but failed to remain accurate in certain anomaly years.

Table 1: Specifications, strengths, weaknesses, and loss function equations of three types of linear regression used

	Type of regularization	Benefits	Drawbacks
Ridge	L2 (add square of coefficients to loss function)	Handles multicollinearity, shrinks coefficients to stabilize estimates	Hard to isolate key features
Lasso	L1 (add absolute value of coefficients to loss function)	Feature selection, isolate important features	Struggles with correlated features
Elastic Net	L1+L2 (combination of both)	Balances ridge and lasso	More complex to tune hyperparameters

$$\text{Ridge: } Loss = \sum (y_i - \hat{y}_i)^2 + \lambda \beta_j^2 \quad (1)$$

$$\text{Lasso: } Loss = \sum (y_i - \hat{y}_i)^2 + \lambda |\beta_j| \quad (2)$$

$$\text{Elastic Net: } Loss = \sum (y_i - \hat{y}_i)^2 + \lambda_1 |\beta_j| + \lambda_2 \beta_j^2 \quad (3)$$

4.3 Time-Series Models

4.3.1 Regression Time-Series Models

To perform the regression time series modeling techniques, yield and all ten feature variables are selected from the dataset and indexed by year. Four types of models were used with average feature data: Differenced Vector AutoRegressive(VAR), AutoRegressive(AR), AutoRegressive Integrated Moving Average(ARIMA), and Seasonal AutoRegressive Integrated Moving Average(SARIMA).

The foundational methodology for time series regression is autoregressive modeling, which forecasts a variable's future values as a linear combination of its previous values determined by a specified lag

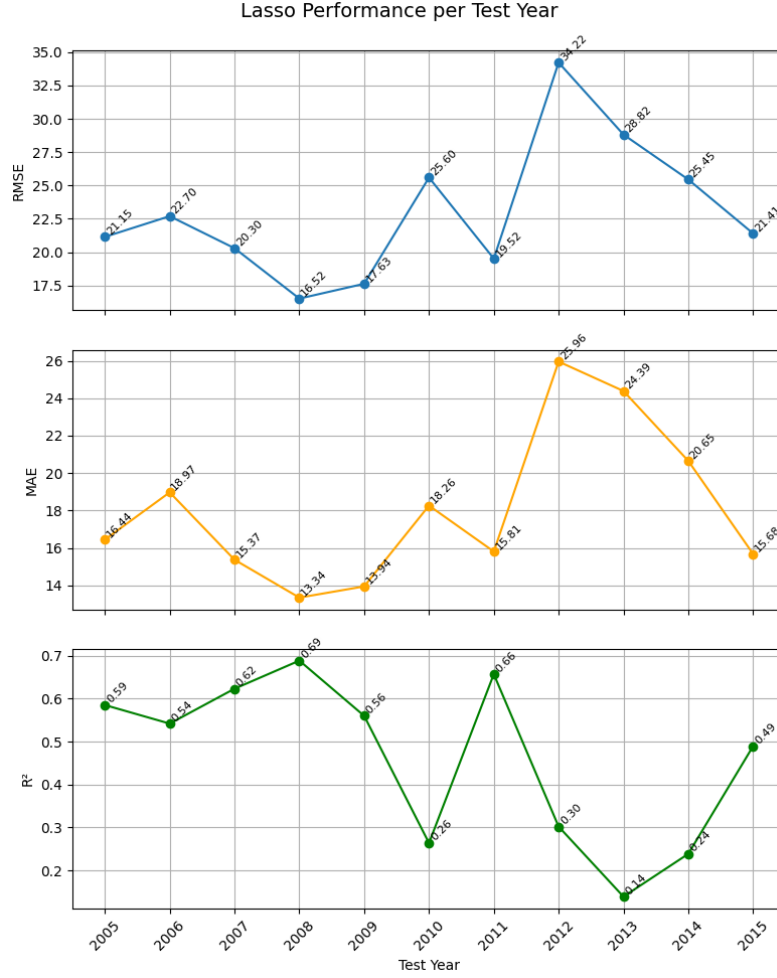


Figure 5: Evaluation metrics for Lasso regression model per test year

order. The AR model is defined as

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \epsilon_t \quad (4)$$

A lag coefficient of 1 was selected for the AR model, meaning that corn yield's future values were forecasted based on its value for the year prior.

VAR modeling extends the AR framework for the analysis of multi-variate time series (Zivot and Wang (2003)). Differencing is applied to VAR modeling to ensure data stationarity. This methodology forecasts corn yield by estimating it through a linear system of its own past values and the past values of feature variables. This model is defined as:

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \end{bmatrix} \quad (5)$$

VAR models assume variables to be stationary, meaning that parameter values (ie, mean, variance) stay constant over time. Non-stationary variables were identified using the Augmented Dickey-Fuller (ADF) test, which checks for a unit root in a time series. The presence of a unit root indicates that a time series' parameters exhibit long-term trends, indicating non-stationarity. First-order differencing was applied to all non-stationary variables before being used to forecast VAR predictions.

ARIMA modeling combines three core components to forecast predictions for a single time series: autoregressive (AR), integrated (I), and moving average (MA). Hyperparameters (p,d,q) specify lag order, differencing order, and moving average order, respectively. Mathematically, the ARIMA model

is defined as:

$$\underbrace{\left(1 - \sum_{i=1}^p \phi_i B^i\right)}_{\text{AR Component}} \underbrace{(1 - B)^d}_{\text{Integrated Component (Differencing)}} Y_t = \mu + \underbrace{\left(1 + \sum_{j=1}^q \theta_j B^j\right)}_{\text{MA Component}} \epsilon_t \quad (6)$$

SARIMA modeling extends the ARIMA structure by enabling it to account for periodic fluctuations within a time series. It does so by introducing three seasonal components, defined as a fourth hyperparameter m . The SARIMA model additionally implements seasonal differencing to ensure stationarity across periods.

$$\text{SARIMA}(p, d, q) \times (P, D, Q)_m$$

$$\underbrace{\left(1 - \sum_{i=1}^p \phi_i B^i\right)}_{\text{Non-Seasonal AR}} \underbrace{\left(1 - \sum_{k=1}^P \Phi_k B^{mk}\right)}_{\text{Seasonal AR}} \underbrace{(1 - B)^d}_{\text{Regular Diff.}} \underbrace{(1 - B^m)^D}_{\text{Seasonal Diff.}} Y_t = \underbrace{\left(1 + \sum_{j=1}^q \theta_j B^j\right)}_{\text{Non-Seasonal MA}} \underbrace{\left(1 + \sum_{l=1}^Q \Theta_l B^{ml}\right)}_{\text{Seasonal MA}} \epsilon_t \quad (7)$$

4.3.2 Deep Learning: LSTM

LSTM + Attention (Long Short Term Memory) is a deep learning model which can "remember" past inputs over long periods of time, making it well suited for time series prediction. The attention mechanism allows the model to selectively focus on different parts of past input.

We used a learning rate of 0.001, dropout rate of 0, and 1 hidden layer. Every model trained with 100 epochs and followed the LOYO cross evaluation strategy.

4.4 Results

Table 2 contains RMSE and MAE metrics for Ridge, Lasso, Elastic Net, SVR (Linear), Differenced VAR, and LSTM + Attention models.

Table 2: Model evaluation metrics for classical regression and time-series models

Model	RMSE	MAE	R^2
Classical Regression Techniques			
Baseline	32.46	25.98	0.28
Ridge	23.93	18.34	0.54
Lasso	23.55	18.06	0.56
Elastic Net	24.00	18.40	0.54
SVR (Linear)	23.94	18.19	0.54
Time Series Techniques			
Baseline	15,432.77	15,432.77	
Differenced VAR	35.41	28.78	-
LSTM + Att	40.53	33.66	-

The baseline result for the classical regression techniques was a linear regression, and the baseline for the time series was the Autoregressive model.

5 Data Communication

For this project, we decided that a dashboard would be an intuitive and dynamic way to showcase this data and explain our results. This dataset yields many possibilities in that there are many combinations of year, county, and feature. Dashboards allow users to navigate these possibilities with the click of a button. Additionally, with the variety of potential parties who may be interested in this work, customization is important. Our application delivers on this with the model-builder tab, as users can train their own models with feature sets of their choice. So, we created this dashboard because it permits dynamic spatiotemporal visualization, while also catering to a diverse audience. As stated in

the introduction, this project has a broad scope. Lawmakers, farmers, traders, and researchers could all benefit from the ability to explore and make actionable predictions with this data.

Users can gain many insights with this dashboard. First, it enables them to explore historical trends. For any county, they can view the mean yield for any year in this time frame. They may also view the coefficient of variation, giving a sense of the uncertainty surrounding the yield. Second, this tool also gives specific forecasts. Users can build their own models using feature sets that they select. They can then make yield forecasts with these models, as they can input feature values themselves. Third, stakeholders can gauge the impact of severe climate conditions with this tool. They may identify counties that are affected by droughts and heat waves at differing severities, which could be especially helpful for lawmakers in need of emergency planning. For example, if a heat wave is expected to hit Illinois one summer, lawmakers could view which counties tend to take the hardest hit in terms of climate stress on yield. These three branches of information communication take place in different tabs of the dashboard.

In terms of software, we utilized a diverse stack to build this tool. It is deployed on Streamlit, which is a budget-friendly platform that easily integrates with GitHub and other Python libraries. So, the code and datasets used to build the dashboard are housed in a GitHub repository, which easily connects with Streamlit. Among these other Python libraries are Pydeck and Sci-Kit Learn. Pydeck delivers interactive, geospatial visualizations which take the form of choropleths in this dashboard. It integrates seamlessly with Streamlit, as does Sci-Kit Learn. We used this popular library to allow users to train their own Ridge, Lasso, and Elastic Net linear models. We also built a variety of functions to display ALE plots (which we will discuss later) along with density plots.

5.1 Tab 1: Exploration

The first tab of the dashboard is a map-based dive into this dataset. As mentioned before, choropleth visualization is a key component of our tool.

Exploration Controls

The screenshot shows the 'Exploration Controls' section of a dashboard. It includes three dropdown menus: 'Dataset Type' with 'Mean' selected, 'Select Year' with '2018' selected, and 'Select Feature' with 'yield' selected. To the right, there is a 'Users Toggle' section with three bullet points: 'Mean v.s. Coeff. Of Variation', 'Year', and 'Variable (yield, temperature, etc.)'.

Figure 6: Example exploration controls

Here, users may select which feature they would like to plot on the heatmap. They also add their year of choice. Additionally, they may view the mean or the coefficient of variation of the selected variable for the year, giving a sense of both central tendency and uncertainty.

Mean: Yield Map - 2018

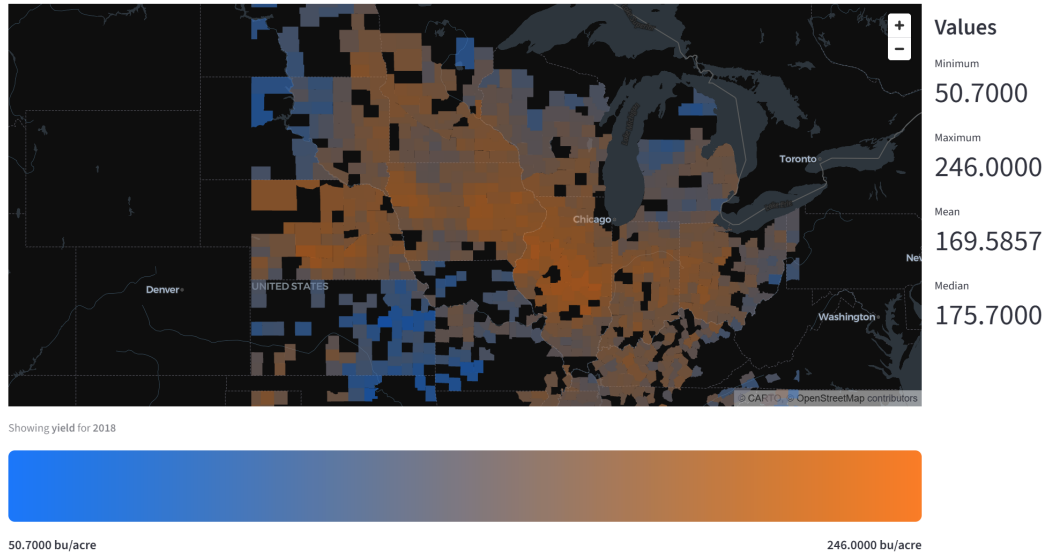


Figure 7: Choropleth output from prior control selections

This is an example choropleth output that shows higher yield throughout the heart of the Corn Belt. Summary statistics are included on the right with a color scale below. This visual is fully interactive, as the user can zoom in and then hover over a county with their cursor to read the county's name and yield. As the controls above are adjusted, this visual shifts almost instantly, allowing for quick analysis of patterns and trends.

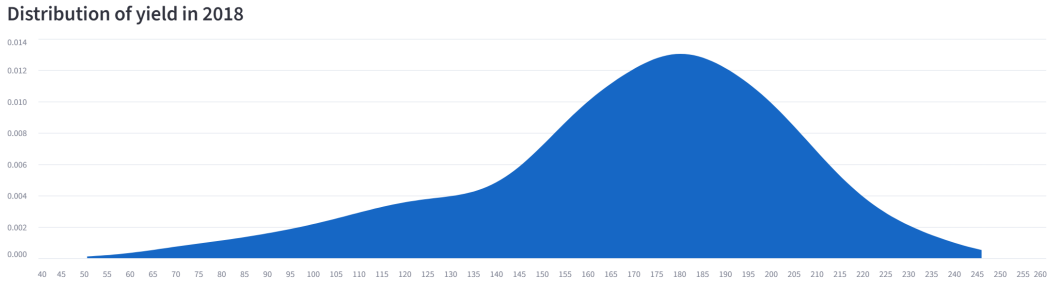


Figure 8: Density plot output from prior control selections

Underneath the choropleth, there is a density plot that shows the spread of the selected variable. As pictured, there was a slight left skew to the mean yield in 2018, although this roughly resembles a normal distribution.

5.2 Tab 2: Climate Stress

The second tab features a customizable metric that we call "climate stress". This variable combines normalized precipitation and maximum temperature to assess the county-level impact of drought and heatwave conditions.

$$\tilde{T}_{i,t} = \frac{T_{i,t} - \min(T_{.,t})}{\max(T_{.,t}) - \min(T_{.,t})}, \quad \tilde{P}_{i,t} = \frac{P_{i,t} - \min(P_{.,t})}{\max(P_{.,t}) - \min(P_{.,t})}. \quad (8)$$

Normalized dryness is defined as

$$D_{i,t} = 1 - \tilde{P}_{i,t}. \quad (9)$$

The climate stress index is given by

$$\text{Stress}_{i,t} = w \tilde{T}_{i,t}^{\alpha} + (1 - w) D_{i,t}^{\beta}$$

(10)

where $\alpha > 0$ and $\beta > 0$ control the nonlinear amplification of heat and moisture stress, respectively, and $w \in [0, 1]$ governs the relative importance of temperature versus precipitation. Thus, users can adjust w if they are more inclined to believe either temperature or precipitation causes more climate stress for a given year or area of interest. They are also able to adjust α and β if there is reason to think that more extreme negative conditions occur during a specific time.

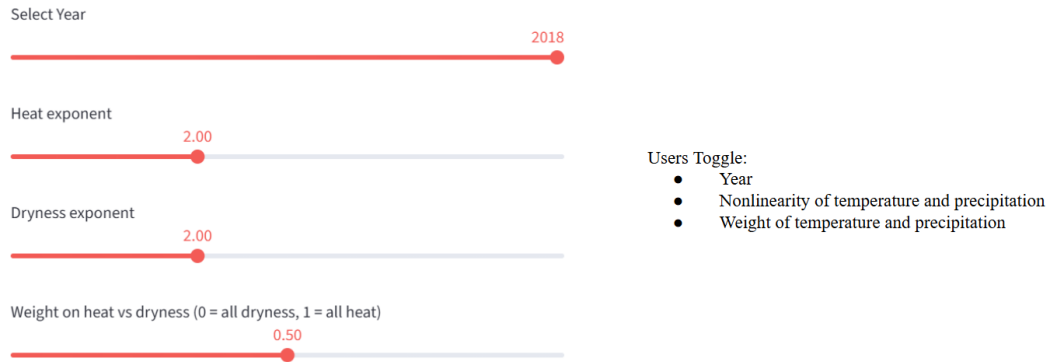


Figure 9: Example climate stress controls

Here, users may control these aforementioned aspects of climate stress calculation. Sliders control both the exponents and the weight. Adjusting the exponents usually gives the stress distribution a right skew, representing abnormal conditions. If users assign more importance to heatwaves, they would move the weight slider right, and vice versa in terms of droughts. They must also select the year they would like to compute climate stress for.

Climate Stress Map – 2018



Figure 10: Choropleth output from prior control selections

Once controls are selected, a choropleth featuring this climate stress metric at a county level is displayed. As shown above, the area that is furthest southwest in the Corn Belt saw the most extreme climate impact. These historical stress patterns could be used by local lawmakers to identify the yield risk counties may see if a drought or heatwave is expected in the near future.

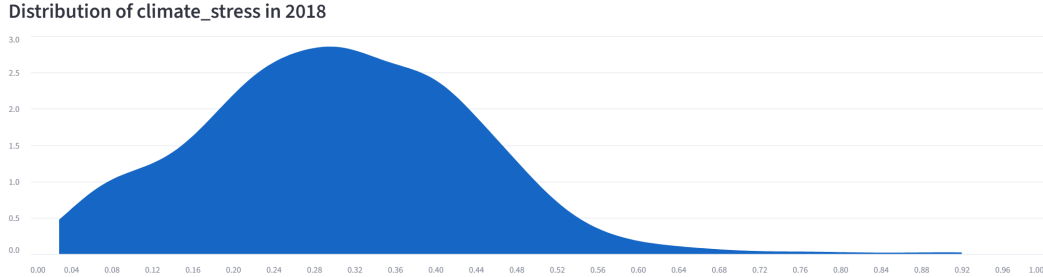


Figure 11: Density plot output from prior control selections

As stated before, exponentially increasing the effect of temperature or dryness causes a right skew. In severe events, some counties may suffer losses far above the rest of the field, and that is seen here.

5.3 Tab 3: Model Builder

Finally, stakeholders can implement their own regression models, allowing for personalized predictive insights. This tab is especially useful in that it gives non-technical parties a way to use forecasting tools they otherwise may not have access to. On top of this, they can deploy these models if they have specific feature values they would like to test.

Model Builder Controls

Select Features for Regression:

tmmx × rmax × pr ×

Select Model Type:

Ridge

Alpha (regularization strength)

1.0000 - +

Users Toggle:

- Feature set
- Model type
- Regularization strength

Figure 12: Example model builder controls

To implement a model, the user must choose a subset of the ten predictive meteorological features contained in this dataset. In terms of model type, they can train three different classical regression models: Ridge, Lasso, and Elastic Net. To tune the models, they can adjust the regularization strength.

Yield Prediction Model (LOYO Ridge)

Predictions are made using leave-one-year-out cross-validation.

Model Performance (LOYO)

	year	rmse	mae
0	1980	47.8265	40.3746
1	1981	30.8464	25.5685
2	1982	29.0782	22.9021
3	1983	52.8503	48.0082
4	1984	36.0173	28.6319
5	1985	28.9809	22.1132
6	1986	28.7974	23.2503
7	1987	26.0367	20.8953
8	1988	52.8294	48.0357
9	1989	33.0416	25.7014

Figure 13: Model performance from prior control selections

As is the case in other parts of the paper, leave-one-year-out cross-validation is utilized to assess model performance. Above is an example performance output that reports RMSE and MAE when each year is left out as the validation set. In other words, users may see how accurate the model is at forecasting yield for each year in the dataset, providing insights as to which years may be most unpredictable.

Predict Your Scenario

Enter values for the selected features. The model will predict yield (bu/acre).

tmmx (K)

297.7914

rmax (%)

90.9211

pr (kg/m²)

3.0004

Predict yield

Figure 14: Form where users can enter their own model inputs

After the user trains their model, they may apply it to real-world scenarios. In this example, the user selected maximum temperature, maximum humidity, and precipitation as their feature set. Once their model generates coefficients, they may input different values for each variable, click "Predict yield", and see the predicted yield from those inputs.

Feature Effect (1D ALE)



Figure 15: ALE plot generator

To gauge feature importance, a dynamic ALE plot generator resides beneath the scenario form. Accumulated Local Effect (ALE) graphs show feature-independent impacts on yield predictions. By binning the feature values, iteratively replacing each bin with extreme values, and evaluating the resulting prediction changes, this method isolates the impact of changes in our climate variables on yield forecasts.

With this dashboard tool, users may select a variable and view how it would be expected to impact yield prediction as this feature value increases. In this example, you can see that as the max temperature increases, the yield predictions from this model decrease.

5.4 Tab 4: Data Dictionary

Data Dictionary

Climate & Environmental Variables

sph — Near-Surface Specific Humidity (g/kg)

Amount of water vapor contained in the air.

vpd — Mean Vapor Pressure Deficit (kPa)

Measures the "drying power" of air — higher VPD = more evaporative demand.

Figure 16: First two entries of the Data Dictionary tab

A data dictionary with descriptions and units is housed in the fourth tab.

Here's a link to the dashboard!

6 Conclusions and Future Work

We introduce an intuitive visual tool that focuses on data insights, yearly feature variations that affect yield, and interactive modeling for yield forecasting. This dashboard has the potential to serve users of a variety of backgrounds, regardless of technical skill level. Stakeholders may explore the data, adjust a custom climate stress metric, and develop personalized yield forecasts.

Our modeling exploration showed that classical regression yielded the best results, particularly from the Lasso regression model. However, these models are limited by their inaccuracy in predicting yield for anomaly years (e.g., droughts) and periods following anomaly years, and future work should aim to address this limitation. Additionally, we struggled with data leakage from the test year in the training set, which resulted in inaccurate metrics. After addressing this issue, we obtained results similar to, though slightly better than, the baseline. The performance of the models represents the optimized configuration to the best of our knowledge. In the future, we plan to address parameter optimization, run diagnostics, and validate each year's performance in greater depth.

Of the four regressive time series models tested, only differenced VAR modeling was able to produce robust results. VAR modeling is the only technique that is able to estimate multivariate systems, which is crucial for crop yield prediction. AR and ARIMA/SARIMA models are univariate functions that can forecast crop yield based on its own past values or the past values of one feature variable. However, crop yield prediction is inherently dependent on multiple weather patterns and therefore, cannot be accurately forecasted without taking all feature data into consideration.

While Siarni-Namini et al. (2018) and Liu et al. (2022) suggests that the LSTM + Attention model should outperform other models, we found that it performed the worst overall.

We concluded that the classical regression models were superior to the more complex models due to the limited temporal sample size. The LSTM and time-series models were likely attempting to learn from the temporal diversity of climate patterns over the test year, but the limited sample size

resulted in overfitting. The classical regression models were able to better handle this limited data by averaging out noise across counties and learning marginal effects, ultimately resulting in smoother extrapolation and avoiding overfitting.

Dashboarding was a relatively smooth process, as Streamlit handled every desired addition well. The only hiccups would come when adding tabs once the tool was already built, as some code would have to be adjusted.

In the future, we aim to search for additional features, potentially related to satellite imagery, terrain information, and other factors that can improve model results. Also, we would like to add granularity to the dashboard tool with county-level soil and pesticide information. Lastly, additional modeling techniques could prove beneficial and achieve more predictive power. With adequate computing strength, one could train a model on daily rather than yearly data. Implementing a distribution regression model would make more sense with these daily entries, and it would likely surpass our current model performance.

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